

NEBULA

Constraint optimization through RTL enhancement using
Generative AI

Final technical report

14 September 2026

Verified result: cycle-exact equivalence plus matched post-route
timing and PPA comparison

Executive summary

Nebula closes the gap between an AI suggestion and an auditable RTL result.

Nebula is a verification-gated RTL optimization workflow. It analyzes timing, maps critical endpoints back to source, supplies bounded context to a Generative AI model, applies structured RTL patches, runs synthesis and timing screens, proves cycle-exact whole-design equivalence, and accepts a candidate only after a matched physical-flow comparison.

The final **ethmac5** demonstration meets the structural benchmark targets: five independent asynchronous master clock domains, one generated clock per master, explicit clock-domain crossings, fixed and programmable divider logic, and approximately 50,000 standard cells. Its frozen baseline contains 51,359 cells after standalone synthesis and 57,608 functional standard cells after routing.

Routed metric	Baseline	Candidate	Change
Setup WNS	-0.430501 ns	-0.335711 ns	+0.094790 ns (+22.019%)
Setup TNS	-22.1458 ns	-18.6490 ns	+3.4968 ns (+15.790%)
Standard cells	57,608	57,508	-100 (-0.174%)
Cell area	139,496 μm^2	139,354 μm^2	-142 μm^2 (-0.102%)
Estimated power	0.230906 W	0.228874 W	-0.002032 W (-0.880%)
Final route DRC	0	0	unchanged

The accepted AI proposal replaces a seven-level linear 32-bit XOR reduction with a balanced three-level tree. EQY proved the complete design cycle-exact and reported PASS after 140 seconds. Both physical layouts completed with zero final route DRC and zero antenna violations.

Claim boundary: timing improves but does not close. Candidate setup WNS remains -0.335711 ns and hold WNS is -0.00689157 ns. Power is a matched vectorless estimate. The frequency figure is a WNS-derived estimate, not a sweep.

1. Problem and design goal

Timing repair at RTL requires more than plausible code generation. A change must preserve interfaces, resets, clock-domain behavior, transaction latency, and functional behavior. It must also produce a measured improvement under the same library, constraints, corner, and physical-flow settings.

Nebula treats the language model as a proposal generator inside a deterministic verification pipeline. The model cannot edit protected clock, reset, CDC, or wrapper files. Candidate patches pass schema validation, path and scope checks, synthesis, static timing, formal equivalence, and post-route comparison before acceptance.

2. System architecture

- **Analyze:** validate the manifest and input hashes, synthesize the full design, run OpenSTA, parse violations, and map endpoints to RTL.
- **Propose:** build bounded hierarchy context around the critical module and request a structured patch from the configured model provider.
- **Verify:** apply the patch in isolation, screen full-design synthesis and timing, then prove whole-design cycle-exact EQY.
- **Measure:** run the same OpenROAD physical flow and accept only when explicit QoR guardrails pass.

Automated onboarding can discover RTL sources, infer an unambiguous top, order module dependencies, expand constant-width clock vectors, detect likely resets/CDC/dividers, and create conservative protected/editable boundaries. Timing intent that cannot be inferred safely remains a human review item.

3. Benchmark qualification

ethmac5 wraps an unchanged OpenCores Ethernet MAC with two auxiliary domains. The five asynchronous masters are WB, TX, RX, AUX0, and AUX1. Each master produces a declared divide-by-two generated clock. Protected two-flop toggle synchronizers provide explicit CDC behavior. Fixed divide-by-two blocks and the MAC's programmable MDC divider satisfy the divider requirement.

Requirement	Evidence
Five master clocks	WB, TX, RX, AUX0, AUX1
Generated clocks	Five divide-by-two generated clocks
CDC	Protected two-flop toggle synchronizers
Divider ratios	Fixed divide-by-two plus programmable MDC
Approx. 50k cells	51,359 synth; 57,608 routed stdcells

The auxiliary state width calibrates the design near the requested size. Filler and tap cells are excluded from the functional standard-cell number. This is a locally constructed qualification wrapper around unchanged third-party MAC RTL, not an upstream ethmac configuration. Upstream licensing notices remain in the source archive.

4. AI-proposed RTL change

The baseline used a serial XOR reduction. The accepted proposal adds intermediate XOR pairs and a balanced final reduction. It preserves ports, state, reset behavior, clocks, CDC logic, and cycle latency. Only **rtl/nebula_aux_domain.v** changes.

```
Baseline: (((((((s0 ^ s1) ^ s2) ^ s3) ^ s4) ^ s5) ^ s6) ^ s7)
Candidate: ((s0 ^ s1) ^ (s2 ^ s3)) ^ ((s4 ^ s5) ^ (s6 ^ s7))
```

Three live-model iterations were retained. Candidate 1 passed all gates. Candidate 2 passed the early screen but was rejected for physical QoR regression. Candidate 3 independently reproduced Candidate 1's netlist and physical metrics. The final deliverable uses Candidate 1.

5. Formal verification

EQY compared the full **nebula_ethmac5_top** hierarchy, rather than only the edited module. The equivalence relation is cycle-exact with the same reset sequence. It proved all partitions and reported **DONE (PASS, rc=0)** after 140 seconds. The package includes the full log, proof configuration, PASS marker, parsed result, and exact patch.

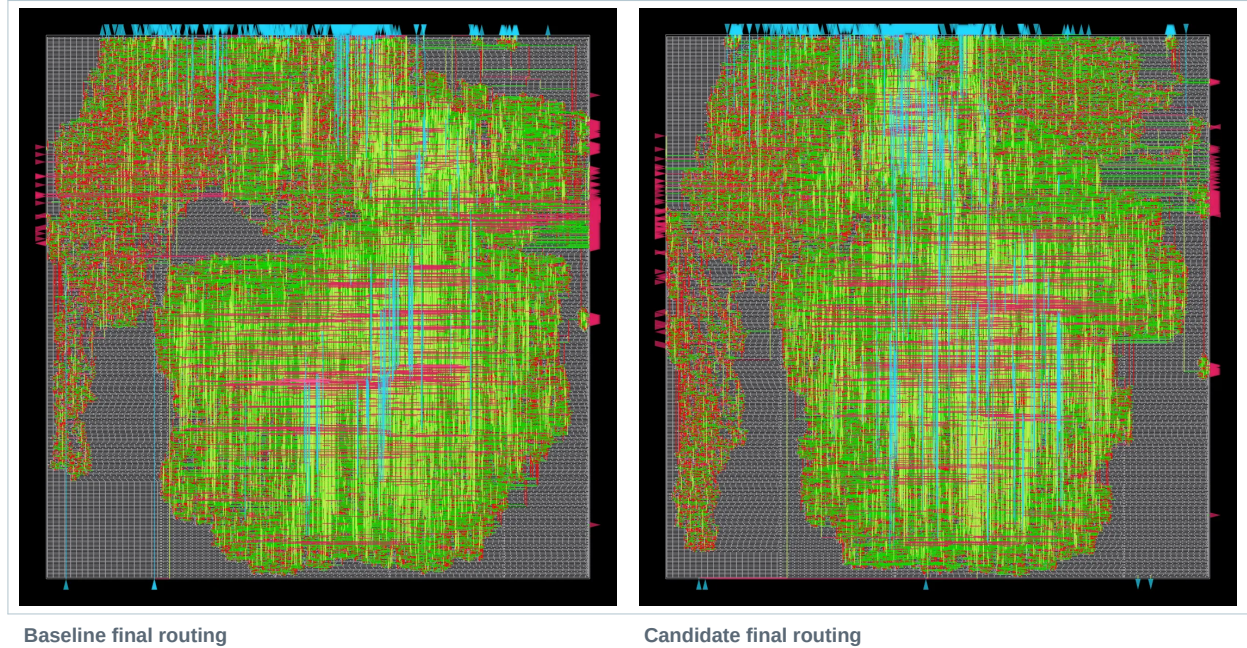
The state machine cannot advance an UNKNOWN or FAIL equivalence result to physical comparison or acceptance. The regression suite exercises this guardrail.

6. Matched physical results

The frozen baseline run is **20260913T160717Z_baseline**. The accepted run is **20260913T164418Z_optimize**, candidate **cand_0001_06c207**. Both use the same toolchain, SDC, Nangate45 library, corner, and physical-flow settings.

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Under the fixed 0.45 ns AUX0 constraint, adding the magnitude of WNS gives an effective-period estimate of 0.880501 ns for the baseline and 0.785711 ns for the candidate, or approximately 1.136 GHz and 1.273 GHz. This 12.1% estimate is explanatory only; it is not a separately measured frequency sweep.

7. Reproducibility and audit trail

The source archive was produced from Git commit **040eb0d**. The proof bundle contains run manifests, locked input hashes, parsed timing data, exact model requests and responses with no credentials, screen results, the raw EQY log, ORFS metadata, final timing reports, layout images, and the acceptance verdict. **SHA256SUMS.txt** covers every distributed file.

Key reproducibility identities

Baseline RTL: e65fb297fde081c698f170e79a277c377929b38b8ab55e51d83b305dcc608443

Accepted RTL: 31c25a0a596d5dc15dd1aa22ddff132954d6734b16b8a7ad3c329f7418a8a0de

Baseline synthesized netlist: f68b90b83507c6c83d44e2382487c233d19fcb03ee72c2694e3259f6a86feb6f

Accepted synthesized netlist: 9e0d8962b27776c3fe0148cbe6ec38df426d95fb09b0e8b85837185308dfc664

Frozen SDC: a71d9f1b8d16fa097a69b722ba79dfb981859d97a4728e27e37fb7bd8a175220

8. Limitations and next work

- Candidate setup WNS remains -0.335711 ns. Candidate hold WNS is -0.00689157 ns and hold TNS is -0.0693946 ns. The result is an optimization, not timing closure.
- Power uses ORFS default vectorless activity. It supports a matched relative comparison, not activity-annotated signoff power.

- The frequency statement derives from fixed-constraint WNS. A separate frequency sweep remains future work.
- Hierarchy context comes from a lightweight source scanner. Complex generate constructs, macros, and ambiguous instance contexts need further validation.

9. Conclusion

Nebula demonstrates a complete evidence chain from timing analysis to an AI proposal, guarded patching, cycle-exact formal proof, and matched post-route measurement. The accepted result improves timing, area, cell count, and estimated power simultaneously while preserving full-design behavior. The retained rejected attempt and explicit limitations show that measured proof, rather than model confidence, controls acceptance.

Appendix A. Proof bundle map

Folder	Contents
formal/	Full EQY log, configuration, PASS marker, parsed result
physical/	Baseline/candidate metadata, finish reports, DRC, antenna and layout images
timing/	Clock report, raw timing, parsed QoR, critical paths and source map
screen/	Candidate pre-formal timing screen
ai/	Three requests, responses and parsed recommendations
comparison/	Iteration history, matched PPA data and verdict
manifests/	Frozen input hashes and run identities
tests/	Regression output and environment summary